Sri Lanka Institute of Information Technology

SE2020 – Web and Mobile Technologies Year 2,Semester 2
Lab Sheet No: 09

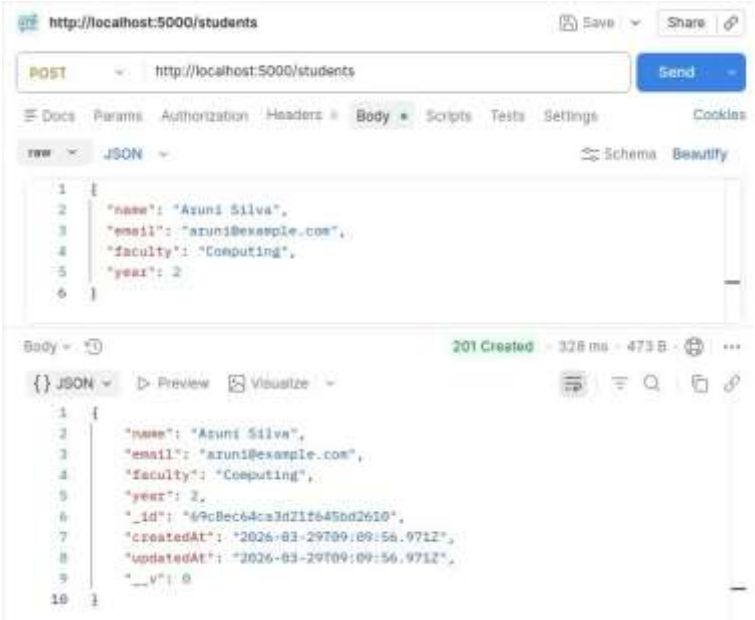
Student ID	IT4100539
Student Name	De Arthur S A
Campus/ Center name	SLIIT KANDY UNI
Specialization	AI
Batch No	01

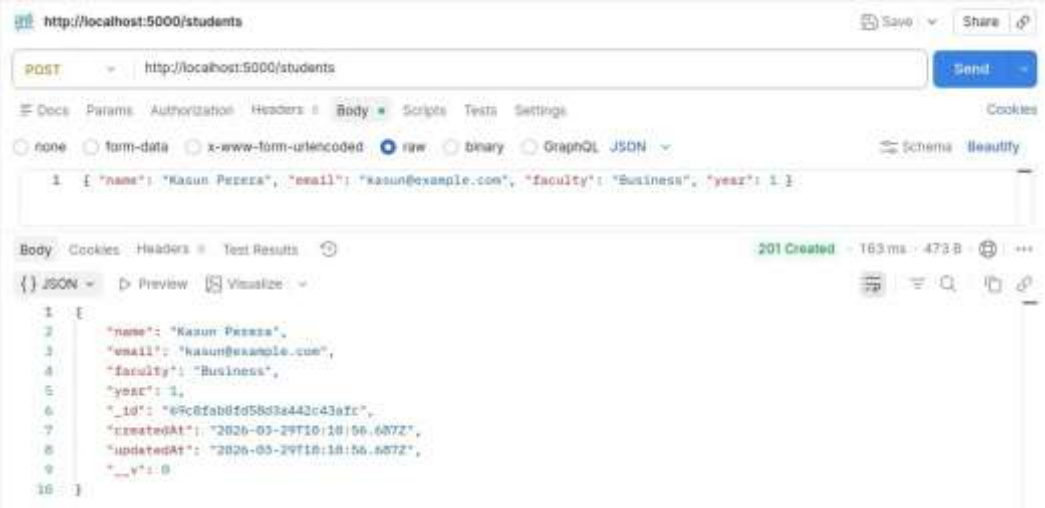
Github URL:
https://github.com/IT24100539/WMT_Lab09.git

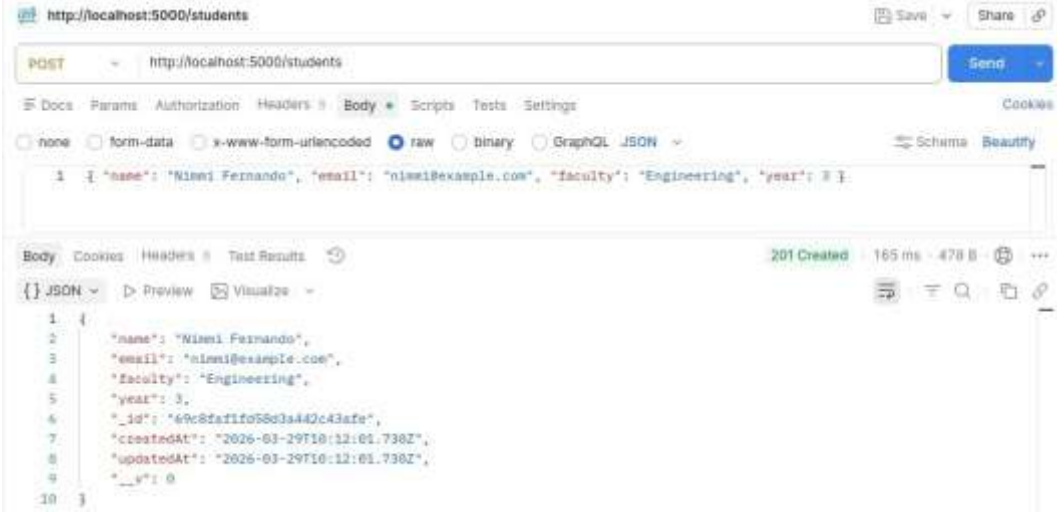
8. Testing with Postman

8.1 Create Initial Data

1. Create at least 3 students.







http://localhost:5000/students/69c8ec64ca3d21f645bd2610

GET http://localhost:5000/students/69c8ec64ca3d21f645bd2610 Send

Docs Params Authorization Headers Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL

This request does not have a body

Body Cookies Headers Test Results 200 OK 222 ms 468 B

JSON Preview Visualize


```
{
    "_id": "69c8ec64ca3d21f645bd2610",
    "name": "Aruni Silva",
    "email": "aruni@example.com",
    "faculty": "Computing",
    "year": 2,
    "createdAt": "2026-03-29T09:09:56.971Z",
    "updatedAt": "2026-03-29T09:09:56.971Z",
    "__v": 0
}
```


2. Create at least 5 menu items.

http://localhost:5000/menu-items

POST http://localhost:5000/menu-items Send

Docs Params Authorization Headers Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify


```
{ "name": "Chicken Rice", "price": 450, "category": "Rice" }
```


Body Cookies Headers Test Results 201 Created 239 ms 464 B

JSON Preview Visualize


```
{
    "name": "Chicken Rice",
    "price": 450,
    "category": "Rice",
    "isAvailable": true,
    "_id": "69c8eea1dcc5aae60f01311a",
    "createdAt": "2026-03-29T09:19:29.090Z",
    "updatedAt": "2026-03-29T09:19:29.090Z",
    "__v": 0
}
```


http://localhost:5000/menu-items

POST http://localhost:5000/menu-items Send

Docs Params Authorization Headers Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify


```
{ "name": "Veggie Burger", "price": 350, "category": "Snack" }
```


Body Cookies Headers Test Results 201 Created 175 ms 466 B

JSON Preview Visualize


```
{
    "name": "Veggie Burger",
    "price": 350,
    "category": "Snack",
    "isAvailable": true,
    "_id": "69c8fccffe58d3a442c43b01",
    "createdAt": "2026-03-29T10:19:59.819Z",
    "updatedAt": "2026-03-29T10:19:59.819Z",
    "__v": 0
}
```

http://localhost:5000/menu-items

Save | Share

POST http://localhost:5000/menu-items Send

Docs | Params | Authorization | Headers | Body | Scripts | Tests | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1 { "name": "Iced Coffee", "price": 150, "category": "Beverage" }
```

Body | Cookies | Headers | Test Results | 201 Created · 376 ms · 467 B

{} JSON | Preview | Visualize


```
1 {
2     "name": "Iced Coffee",
3     "price": 150,
4     "category": "Beverage",
5     "isAvailable": true,
6     "_id": "69c8fd8afd58d3a442c43b03",
7     "createdAt": "2026-03-29T10:20:58.685Z",
8     "updatedAt": "2026-03-29T10:20:58.685Z",
9     "__v": 0
10 }
```


http://localhost:5000/menu-items

Save | Share

POST http://localhost:5000/menu-items Send

Docs | Params | Authorization | Headers | Body | Scripts | Tests | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1 { "name": "Fish Patty", "price": 60, "category": "Snack" }
```

Body | Cookies | Headers | Test Results | 201 Created · 195 ms · 462 B

{} JSON | Preview | Visualize


```
1 {
2     "name": "Fish Patty",
3     "price": 60,
4     "category": "Snack",
5     "isAvailable": true,
6     "_id": "69c8fd35fd58d3a442c43b05",
7     "createdAt": "2026-03-29T10:21:41.924Z",
8     "updatedAt": "2026-03-29T10:21:41.924Z",
9     "__v": 0
10 }
```


http://localhost:5000/menu-items

Save | Share

POST http://localhost:5000/menu-items Send

Docs | Params | Authorization | Headers | Body | Scripts | Tests | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1 { "name": "Mixed Fried Rice", "price": 550, "category": "Rice" }
```

Body | Cookies | Headers | Test Results | 201 Created · 228 ms · 468 B

{} JSON | Preview | Visualize


```
1 {
2     "name": "Mixed Fried Rice",
3     "price": 550,
4     "category": "Rice",
5     "isAvailable": true,
6     "_id": "69c8fd64fd58d3a442c43b07",
7     "createdAt": "2026-03-29T10:22:28.176Z",
8     "updatedAt": "2026-03-29T10:22:28.176Z",
9     "__v": 0
10 }
```

http://localhost:5000/menu-items

Save Share

GET http://localhost:5000/menu-items Send

Docs Params Authorization Headers Body Scripts Tests Settings Cookies

Body Cookies Headers Test Results 200 OK 271 ms 1.21 KB

{} JSON Preview Visualize


```
[
    {
        "_id": "69c8fd64fd58d3a442c43b07",
        "name": "Mixed Fried Rice",
        "price": 550,
        "category": "Rice",
        "isAvailable": true,
        "createdAt": "2026-03-29T10:22:28.176Z",
        "updatedAt": "2026-03-29T10:22:28.176Z",
        "__v": 0
    },
    {
        "_id": "69c8fd35fd58d3a442c43b05",
        "name": "Fish Patty",
        "price": 60,
        "category": "Snack",
        "isAvailable": true,
        "createdAt": "2026-03-29T10:21:41.924Z",
        "updatedAt": "2026-03-29T10:21:41.924Z",
        "__v": 0
    },
    {
        "_id": "69c8fd8afd58d3a442c43b03",
        "name": "Iced Coffee",
        "price": 150,
        "category": "Beverage",
        "isAvailable": true,
        "createdAt": "2026-03-29T10:20:58.605Z",
        "updatedAt": "2026-03-29T10:20:58.605Z",
        "__v": 0
    },
    {
        "_id": "69c8fccffd58d3a442c43b01",
        "name": "Veggie Burger",
        "price": 350,
        "category": "Snack",
        "isAvailable": true,
        "createdAt": "2026-03-29T10:19:59.819Z",
        "updatedAt": "2026-03-29T10:19:59.819Z",
```


Connect Git Console Terminal Globals Vault Tools

3. Place multiple orders.

Order 1 (Aruni):

http://localhost:5000/orders

Save Share

POST http://localhost:5000/orders Send

Docs Params Authorization Headers Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify


```
{
  "student": "69c8ec64ca3d21f645bd2610",
  "items": [{ "menuItem": "69c8eea1dcc5aae60f01311a", "quantity": 2 }]
}
```


Body Cookies Headers Test Results 201 Created · 836 ms · 829 B

{} JSON Preview Visualize


```
{
    "_id": "69c8fed9fd58d3a442c43b0d",
    "student": {
        "_id": "69c8ec64ca3d21f645bd2610",
        "name": "Aruni Silva",
        "email": "aruni@example.com",
        "faculty": "Computing",
        "year": 2,
        "createdAt": "2026-03-29T09:09:56.971Z",
        "updatedAt": "2026-03-29T09:09:56.971Z",
        "__v": 0
    },
    "items": [
        {
            "menuItem": {
                "_id": "69c8eea1dcc5aae60f01311a",
                "name": "Chicken Rice",
                "price": 450,
                "category": "Rice",
                "isAvailable": true,
                "createdAt": "2026-03-29T09:19:29.090Z",
                "updatedAt": "2026-03-29T09:19:29.090Z",
                "__v": 0
            },
            "quantity": 2
        }
    ],
    "totalPrice": 900,
    "status": "PLACED",
    "createdAt": "2026-03-29T10:28:41.389Z",
    "__v": 0
}
```


Connect Git Console Terminal Globals Vault Tools

Order 2 (Kasun):

http://localhost:5000/orders

Save Share

POST http://localhost:5000/orders Send

Docs Params Authorization Headers Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify


```
{
  "student": "69c8fab0fd58d3a442c43afc",
  "items": [{ "menuItem": "69c8fd0afd58d3a442c43b03", "quantity": 3 }, { "menuItem": "69c8fd35fd58d3a442c43b05",
    "quantity": 5 }]
}
```

Docs Params Authorization Headers 8 Body Scripts Tests Settings

Body Cookies Headers 8 Test Results

{} JSON Preview Visualize


```
{
    "_id": "69c8fffffd58d3a442c43b14",
    "student": {
        "_id": "69c8fab0fd58d3a442c43afc",
        "name": "Kasun Perera",
        "email": "kasun@example.com",
        "faculty": "Business",
        "year": 1,
        "createdAt": "2026-03-29T10:10:56.687Z",
        "updatedAt": "2026-03-29T10:10:56.687Z",
        "__v": 0
    },
    "items": [
        {
            "menuItem": {
                "_id": "69c8fd0afd58d3a442c43b03",
                "name": "Iced Coffee",
                "price": 150,
                "category": "Beverage",
                "isAvailable": true,
                "createdAt": "2026-03-29T10:20:58.605Z",
                "updatedAt": "2026-03-29T10:20:58.605Z",
                "__v": 0
            },
            "quantity": 3
        },
        {
            "menuItem": {
                "_id": "69c8fd35fd58d3a442c43b05",
                "name": "Fish Patty",
                "price": 60,
                "category": "Snack",
                "isAvailable": true,
                "createdAt": "2026-03-29T10:21:41.924Z",
                "updatedAt": "2026-03-29T10:21:41.924Z",
                "__v": 0
            },
            "quantity": 5
        }
    ],
    "totalPrice": 750,
    "status": "PLACED",
    "createdAt": "2026-03-29T10:33:35.752Z",
    "__v": 0
}
```

8.2 Functional Testing

1. Verify CRUD operations.

Update – PATCH

http://localhost:5000/orders/69c8fffffd58d3a442c43b14/status

Save | Share

PATCH http://localhost:5000/orders/69c8fffffd58d3a442c43b14/status Send

Docs Params Authorization Headers 8 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
{
    "status": "DELIVERED"
}
```

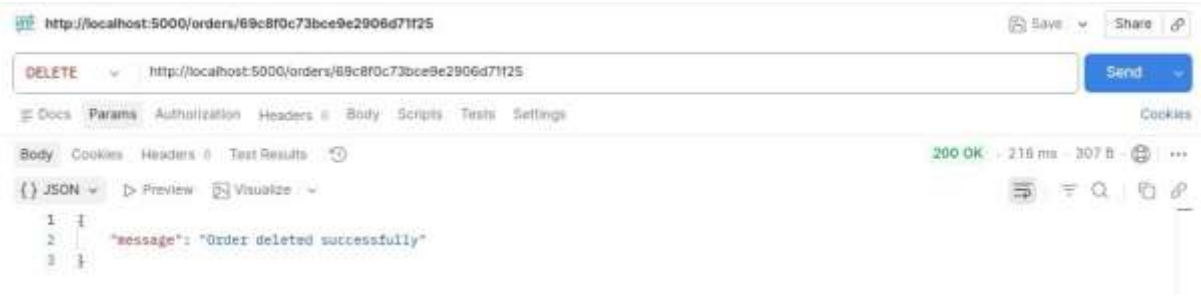
Docs Params Authorization Headers 8 Body Scripts Tests Settings

Body Cookies Headers 8 Test Results

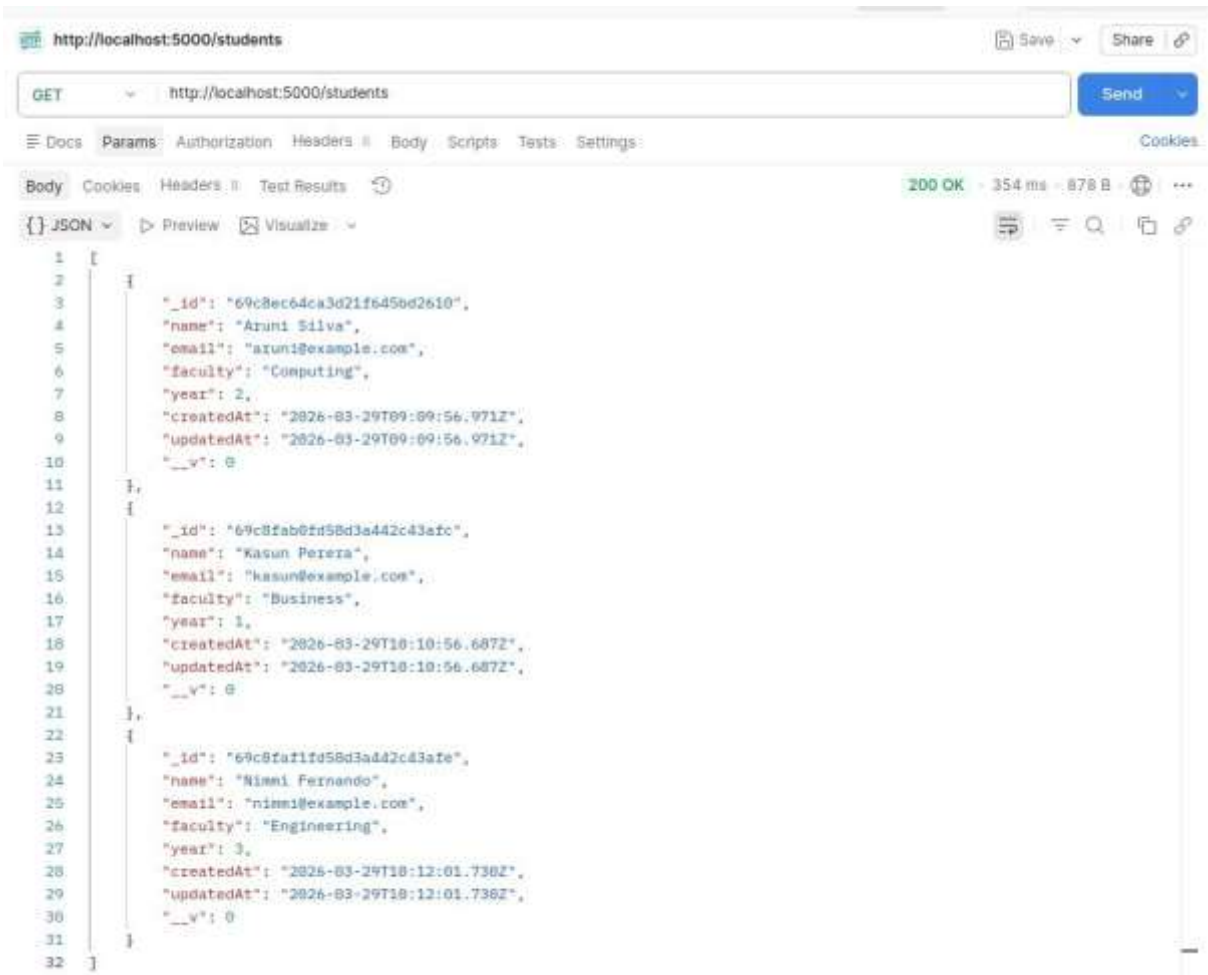
{} JSON Preview Visualize


```
{
    "_id": "69c8fffffd58d3a442c43b14",
    "student": {
        "_id": "69c8fab0fd58d3a442c43afc",
        "name": "Kasun Perera",
        "email": "kasun@example.com",
        "faculty": "Business",
        "year": 1,
        "createdAt": "2026-03-29T10:10:56.687Z",
        "updatedAt": "2026-03-29T10:10:56.687Z",
        "__v": 0
    },
    "items": [
        {
            "menuItem": {
                "_id": "69c8fd0afd58d3a442c43b03",
                "name": "Iced Coffee",
                "price": 150,
                "category": "Beverage",
                "isAvailable": true,
                "createdAt": "2026-03-29T10:20:58.605Z",
                "updatedAt": "2026-03-29T10:20:58.605Z",
                "__v": 0
            },
            "quantity": 3
        },
        {
            "menuItem": {
                "_id": "69c8fd35fd58d3a442c43b05",
                "name": "Fish Patty",
                "price": 60,
                "category": "Snack",
                "isAvailable": true,
                "createdAt": "2026-03-29T10:21:41.924Z",
                "updatedAt": "2026-03-29T10:21:41.924Z",
                "__v": 0
            },
            "quantity": 5
        }
    ],
    "totalPrice": 750,
    "status": "DELIVERED",
    "createdAt": "2026-03-29T10:33:35.752Z",
    "__v": 0
}
```


Delete - DELETE



Read - GET



2. Test search and pagination.

Search :

http://localhost:5000/menu-items/search

Save | Share

GET http://localhost:5000/menu-items/search?name=Rice Send

Docs | Params | Authorization | Headers | Body | Scripts | Tests | Settings | Cookies

Body | Cookies | Headers | Test Results | 200 OK · 190 ms · 658 B

{} JSON | Preview | Visualize


```
[
    {
        "_id": "69c8eea1dcc5aae60f01311a",
        "name": "Chicken Rice",
        "price": 450,
        "category": "Rice",
        "isAvailable": true,
        "createdAt": "2026-03-29T09:19:29.090Z",
        "updatedAt": "2026-03-29T09:19:29.090Z",
        "__v": 0
    },
    {
        "_id": "69c8fde4fd58d3a442c43b07",
        "name": "Mixed Fried Rice",
        "price": 550,
        "category": "Rice",
        "isAvailable": true,
        "createdAt": "2026-03-29T10:22:28.176Z",
        "updatedAt": "2026-03-29T10:22:28.176Z",
        "__v": 0
    }
]
```

Pagination :

http://localhost:5000/orders

Save Share

GET http://localhost:5000/orders?page=1&limit=2 Send

Docs Params Authorization Headers Body Scripts Tests Settings Cookies

Body Cookies Headers 8 Test Results

{} JSON Preview Visualize


```
{
    "totalOrders": 1,
    "currentPage": 1,
    "totalPages": 1,
    "orders": [
        {
            "_id": "69c8fffffd58d3a442c43b14",
            "student": {
                "_id": "69c8fab0fd58d3a442c43afc",
                "name": "Kasun Perera",
                "email": "kasun@example.com",
                "faculty": "Business",
                "year": 1,
                "createdAt": "2026-03-29T10:10:56.687Z",
                "updatedAt": "2026-03-29T10:10:56.687Z",
                "__v": 0
            },
            "items": [
                {
                    "menuItem": {
                        "_id": "69c8fd0afd58d3a442c43b03",
                        "name": "Iced Coffee",
                        "price": 150,
                        "category": "Beverage",
                        "isAvailable": true,
                        "createdAt": "2026-03-29T10:20:58.605Z",
                        "updatedAt": "2026-03-29T10:20:58.605Z",
                        "__v": 0
                    },
                    "quantity": 3
                },
                {
                    "menuItem": {
                        "_id": "69c8fd35fd58d3a442c43b05",
                        "name": "Fish Patty",
                        "price": 60,
                        "category": "Snack",
                        "isAvailable": true,
                        "createdAt": "2026-03-29T10:21:41.924Z",
                        "updatedAt": "2026-03-29T10:21:41.924Z",
                        "__v": 0
                    },
                    "quantity": 5
                }
            ],
            "totalPrice": 750,
            "status": "DELIVERED",
            "createdAt": "2026-03-29T10:33:35.752Z",
            "__v": 0
        }
    ]
```


Connect Git Console Terminal

Validate Aggregation

Total Spent:

http://localhost:5000/analytics/total-spent/69c8fab0fd58d3a442c43afc

GET http://localhost:5000/analytics/total-spent/69c8fab0fd58d3a442c43afc

Send

Docs Params Authorization Headers Body Scripts Tests Settings Cookies

Body Cookies Headers Test Results 200 OK 220 ms 324 B

JSON Preview Visualize


```
{
    "studentId": "69c8fab0fd58d3a442c43afc",
    "totalSpent": 750
}
```


Top Selling Items:

http://localhost:5000/analytics/top-menu-items

GET http://localhost:5000/analytics/top-menu-items

Send

Docs Params Authorization Headers Body Scripts Tests Settings Cookies

Body Cookies Headers Test Results 200 OK 167 ms 783 B

JSON Preview Visualize


```
[
    {
        "_id": "69c8fd35fd58d3a442c43b05",
        "totalQuantity": 5,
        "menuItem": {
            "_id": "69c8fd35fd58d3a442c43b05",
            "name": "Fish Patty",
            "price": 60,
            "category": "Snack",
            "isAvailable": true,
            "createdAt": "2026-03-29T10:21:41.924Z",
            "updatedAt": "2026-03-29T10:21:41.924Z",
            "__v": 0
        }
    },
    {
        "_id": "69c8fd8afd58d3a442c43b03",
        "totalQuantity": 3,
        "menuItem": {
            "_id": "69c8fd8afd58d3a442c43b03",
            "name": "Iced Coffee",
            "price": 150,
            "category": "Beverage",
            "isAvailable": true,
            "createdAt": "2026-03-29T10:20:58.605Z",
            "updatedAt": "2026-03-29T10:20:58.605Z",
            "__v": 0
        }
    }
]
```